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EC5230 - Industrial Organisation

Lecture 7 - Bundling and Tying

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Introduction

This lecture

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Core question: When and why do firms bundle products together?

Learning objectives:

1. Understand bundling strategies under monopoly (separate, pure, mixed)
2. Analyze bundling as entry deterrence mechanism
3. Evaluate bundling as product differentiation tool
4. Apply welfare analysis to bundling decisions

Key tools: Price discrimination, consumer valuations, strategic interaction

What is bundling?

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Definition: Selling two or more products/services in a single package

Tying: A special case

What is bundling?

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often used for complementary goods

Why study bundling?

- Extracts consumer surplus through price discrimination
- Alters competitive dynamics (entry deterrence)
- Creates product differentiation in homogeneous markets

Key mechanism: Exploiting heterogeneity in consumer valuations (reservation prices)

Why Bundling Matters: The Surplus Extraction Mechanism

Core insight: Bundling raises profits even under independent valuations; gains are largest when valuations are negatively correlated.

Suppose Consumer 1 values A highly but B little, and Consumer 2 is the opposite.

Without bundling: Must set uniform prices; some consumers buy nothing

With bundling: Package the goods; consumers with diverse preferences both buy

Result: Monopolist extracts surplus from more consumers at higher average price

This is the key driver behind all three bundling motives we study today.

Bundling Under Monopoly

Bundling under monopoly: firm strategies

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Assume a monopoly produces two goods in the market: A and B.

- Separate selling
  - $\rightarrow$ A and B sold individually (at prices p_A and p_B)
  - $\rightarrow$ Buying one A and one B costs $p_A + p_B$
- Pure bundling
  - $\rightarrow$ A and B sold only in a bundle, not individually
  - $\rightarrow$ Buying a bundle consisting of one A and one B costs p_{AB}
- Mixed bundling
  - $\rightarrow$ A and B sold both individually and in a bundle
  - $\rightarrow$ Three prices set: p_A, p_B, and p_{AB}

Bundling under monopoly: consumer valuations

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Assume each consumer has independent willingness-to-pay for A and B:

$$\theta_A \sim U[0,1], \quad \theta_B \sim U[0,1]$$

Combined valuation:

$$\Theta = \theta_A + \theta_B$$

Consumer purchases:

- A only if $\theta_A \geq p_A$ and $\theta_B < p_{AB} - p_A$
- B only if $\theta_B \geq p_B$ and $\theta_A < p_{AB} - p_B$
- Bundle AB if $\Theta \geq p_{AB}$
- Nothing if none of the above

Bundling under monopoly: demand under separate selling

Source: Belleflamme and Peitz (2015), Figure 11.2

Bundling under monopoly: demand under mixed bundling

Source: Belleflamme and Peitz (2015), Figure 11.4

Bundling under monopoly: optimal prices and profits

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Assume: Marginal costs = 0, symmetric prices $p_A = p_B = p$ where applicable

Separate selling

$$\text{Demand:} \quad D_A = D_B = 1 - p$$

$$\text{Profit:} \quad \pi^{\text{sep}} = 2p(1-p)$$

$$\text{Optimal:} \quad p^{\text{sep}} = \frac{1}{2} \quad \Rightarrow \quad \pi^{\text{sep}} = \frac{1}{2}$$

Bundling under monopoly: pure vs. mixed bundling

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Pure bundling

Combined demand follows from $\Theta \sim \text{Triangular } [0, 2]$:

$$D_{AB}^{\text{pure}} = 1 - \frac{1}{2} p_{AB}^2$$

$$\text{Optimal: } \; p_{AB}^{\text{pure}} = \sqrt{\frac{2}{3}} \approx 0.82 \quad \Rightarrow \quad \pi^{\text{pure}} \approx 0.544$$

Mixed bundling (with $p_A = p_B = p$)

Firm maximizes over both p and p_{AB}:

$$\text{Optimal: } \; p^{\text{mix}} \approx 0.67, \quad p_{AB}^{\text{mix}} \approx 0.86 \quad \Rightarrow \quad \pi^{\text{mix}} \approx 0.549$$

Bundling under monopoly: profit ranking

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Strategy	Individual price	Bundle price	Profit
Separate	0.50	–	0.500
Pure bundle	–	0.82	0.544
Mixed	0.67	0.86	0.549

! Important

Mixed bundling yields highest profit; pure bundling dominates separate selling.

Bundling under monopoly: welfare analysis

Source: Belleflamme and Peitz (2015), Figure 11.5

Bundling under monopoly: welfare ranking

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Producer surplus: Mixed bundling highest (monopolist extracts most surplus)

Consumer surplus: Pure bundling highest (though prices rise, more consumers are served)

Total welfare (Belleflamme and Peitz 2015, Ch. 11):

$$W^{\text{pure}} > W^{\text{mix}} > W^{\text{sep}}$$

Note

Intuition: Pure bundling reduces effective price discrimination distortion by serving broader demand.

Bundling under monopoly: worked example

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Setup: Three consumers, valuations:

Consumer	θ_A	θ_B	Θ
1	10	0	10
2	7.5	7.5	15
3	0	10	10

Assume $MC = 0$.

Bundling under monopoly: worked example — separate selling

Set $p_A = p_B = 7.5$:

- Consumer 1: Buys A (surplus: $10 - 7.5 = 2.5$)
- Consumer 2: Buys both (surplus: $15 - 15 = 0$)
- Consumer 3: Buys B (surplus: $10 - 7.5 = 2.5$)

$$\pi^{\text{sep}} = \underbrace{2 \times 7.5}_{\text{good A}} + \underbrace{2 \times 7.5}_{\text{good B}} = 30$$

Bundling under monopoly: worked example — pure bundling

Set $p_{AB} = 10$:

- All consumers buy bundle (since $\Theta \in \{10, 15, 10\}$)

$$\pi^{\text{pure}} = 3 \times 10 = 30$$

Bundling under monopoly: worked example — mixed bundling

Set $p_A = p_B = 10, p_{AB} = 15$:

- Consumer 1: Buys A only (surplus: 0)
- Consumer 2: Buys bundle (surplus: 0)
- Consumer 3: Buys B only (surplus: 0)

$$\pi^{\text{mix}} = 10 + 10 + 15 = 35$$

! Important

Mixed bundling extracts all consumer surplus and yields highest profit.

Bundling under monopoly: exercise

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Setup: Same valuations as before, but now $MC = 6$.

Task: Compute optimal prices and profits under all three strategies.

Hint: Remember to compute profit as $(p - MC) \times Q$. Some consumers may not purchase if the price exceeds their willingness to pay.

Bundling as Entry Deterrence

Bundling as entry deterrence: strategic motivation

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Setting: Incumbent monopolist (Firm 1) in market A seeks to deter entry in market B

Mechanism:

1. Firm 1 bundles A with B at attractive price
2. Entrant (Firm 2) would enter B at separate prices
3. Bundling reduces Firm 2's residual demand in B
4. If bundling drives Firm 2's profit below entry cost F, entry is deterred

Key trade-off: Firm 1 sacrifices profit in B (short term) to monopolize it (long term)

Bundling as entry deterrence: Cournot competition

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Setup:

- Market A: Firm 1 has monopoly, demand $p_A = 1 - q_A$
- Market B: Potential duopoly, demand $p_B = 1 - q_{B1} - q_{B2}$
- Firm 1 can bundle; Firm 2 enters market B or stays out
- $MC_1 = MC_2 = 0$, entry cost F

Without bundling (separate pricing):

$$\pi_1^{\text{sep}} = \underbrace{\frac{1}{4}}_{\text{A monopoly}} + \underbrace{\frac{1}{9}}_{\text{B Cournot}} = \frac{13}{36} \approx 0.361, \quad \pi_2^{\text{sep}} = \frac{1}{9} \approx 0.111$$

With bundling (Belleflamme and Peitz 2015, Ch. 11):

Bundling as entry deterrence: Cournot competition

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$$\pi_1^{\text{bun}} = \frac{18}{49} \approx 0.367, \quad \pi_2^{\text{bun}} = \frac{4}{49} \approx 0.082$$

Bundling as entry deterrence: entry deterrence condition

Entry deterrence window:

If $F \in \left(\frac{4}{49}, \frac{1}{9}\right)$ or approximately $F \in (0.082, 0.111)$:

- Without bundling: Firm 2 enters (earns $\pi_2^{\text{sep}} - F > 0$)
- With bundling: Firm 2 stays out (earns $\pi_2^{\text{bun}} - F < 0$)

Firm 1's calculation:

- Duopoly profit without deterrence: $\frac{13}{36} \approx 0.361$
- Monopoly profit with deterrence: $\frac{18}{49} \approx 0.367$

Since $0.367 > 0.361$, bundling is a profitable deterrence strategy.

Bundling as entry deterrence: Bertrand competition

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Setup:

- Firm 1 monopolist in A; Firm 2 is potential entrant in B
- Consumers have unit demand for one bundle (A + B required for utility)
- Valuations: $\theta_A, \theta_B \sim U[0,1]$ independent
- Entry cost: F

Benchmark (no Firm 1 entry in B):

$$p_A = p_B = 0.5, \quad \pi_1 = \pi_2 = 0.25$$

With Firm 1 also competing in B via separate prices (no bundling):

$$\pi_2^{\text{sep}} \approx 0.12$$

Firm 2 enters if $F < 0.12$.

Bundling as entry deterrence: Bertrand — bundling strategy

With bundling: Firm 1 offers bundle at aggressive price

- Firm 2's residual profit: $\pi_2^{\text{bun}} \approx 0.067$
- Firm 1's bundling profit: $\pi_1^{\text{bun}} \approx 0.37$

Entry deterrence window: $F \in (0.067, 0.12)$

- Without bundling (Firm 1 competes in B at separate prices): Firm 2 enters (earns $\approx 0.12 > F$)
- With bundling: Firm 2 stays out (earns $\approx 0.067 < F$)
- Firm 1 prefers bundling: earns ≈ 0.37 vs. ≈ 0.25 (A-only benchmark)

Note

Numerical results from Belleflamme and Peitz (2015), Chapter 11.

Bundling as entry deterrence: welfare implications

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Consumer perspective:

- Bundling monopolizes market B $\longrightarrow$ higher prices
- Loss of competition reduces consumer choice

Efficiency defence:

- Avoids duplication of entry costs

Antitrust concern:

- Leveraging market power (A) into market B via bundling
- Exclusionary practice that harms consumer welfare
- Yet efficiency defense possible if entry costs substantial

Bundling as Product Differentiation

Bundling as product differentiation: motivation

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Problem: Bertrand duopoly with homogeneous goods yields $\pi = 0$

Solution via bundling: Tie product to complementary service

- Creates vertical product differentiation
- Softens price competition
- Allows positive equilibrium profits

Example: Computer + tech support, airline ticket + lounge access, software + cloud storage

Bundling as product differentiation: the Shy (1995) model

Source: Shy (1995), Ch. 5

Competitors:

- Firm 1: Sells product bundled with service at price p_S
- Firm 2: Sells product only at price p_N

Cost structure:

- Product production: $MC_p = m > 0$ (same for both)
- Service provision: $MC_s = w \in (0, 2)$ (Firm 1 only)

Consumer valuations:

$$U_S = B - p_S + s, \quad U_N = B - p_N$$

Bundling as product differentiation: the Shy (1995) model

where $s \sim U[0,1]$ is idiosyncratic service valuation, $B > 0$ is the common base product value.

Assume: B sufficiently large that all consumers purchase in equilibrium (full market coverage).

Bundling as product differentiation: equilibrium indifference

Indifferent consumer $\hat{s}$:

Consumer buys bundle if $s > \hat{s}$, where:

$$B - p_S + \hat{s} = B - p_N \quad \Rightarrow \quad \hat{s} = p_S - p_N$$

Demands:

$$D_S = 1 - \hat{s} = 1 - (p_S - p_N)$$

$$D_N = \hat{s} = p_S - p_N$$

Bundling as product differentiation: best response functions

Firm 1 (bundled):

$$\max_{p_S}(p_S - m - w)(1 - p_S + p_N)$$

$$BR_1: \quad p_S = \frac{1}{2}(1 + m + w + p_N)$$

Firm 2 (unbundled):

$$\max_{p_N}(p_N - m)(p_S - p_N)$$

$$BR_2: \quad p_N = \frac{1}{2}(m + p_S)$$

Bundling as product differentiation: equilibrium prices and profits

Solving the system (overbars denote Nash equilibrium values throughout):

$$\bar{p}_S = \frac{2}{3}(1+w) + m, \quad \bar{p}_N = \frac{1}{3}(1+w) + m$$

Equilibrium profits:

$$\bar{\pi}_S = \frac{(2-w)^2}{9}, \quad \bar{\pi}_N = \frac{(1+w)^2}{9}$$

Both are positive for all $w \in (0, 2)$

Bundling as product differentiation: equilibrium prices and profits

Bundling overcomes the Bertrand paradox.

Bundling as product differentiation: comparative statics

How does service cost w affect equilibrium?

Prices:

$$\frac{\partial \bar{p}_S}{\partial w} = \frac{2}{3} > 0, \quad \frac{\partial \bar{p}_N}{\partial w} = \frac{1}{3} > 0$$

Both firms raise prices as service becomes more costly

Service penetration (equilibrium threshold $\bar{s} \equiv \hat{s}^*$):

$$\bar{s} = \bar{p}_S - \bar{p}_N = \frac{1+w}{3}$$

As w rises, $\bar{p}_S$ increases by $\frac{2}{3}$ per unit vs. $\frac{1}{3}$ for $\bar{p}_N$: the price gap widens, raising $\bar{s}$ and shrinking bundle demand.

Bundling as product differentiation: social optimum

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First-best: Serve consumer s if $s \geq w$ (service cost)

$$s^* = w$$

Market equilibrium:

$$\bar{s} = \frac{1+w}{3}$$

Provision distortion:

- If $w < \frac{1}{2}$: $\bar{s} > s^*$ $\longrightarrow$ too few consumers buy the bundle (market threshold too high)
- If $w > \frac{1}{2}$: $\bar{s} < s^*$ $\longrightarrow$ too many consumers buy the bundle (market threshold too low)

Bundling as product differentiation: social optimum

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Note

Competitive bundling still distorts service allocation, but may be welfare-improving relative to the Bertrand equilibrium without bundling.

Bundling as product differentiation: welfare comparison

Three scenarios:

1. Bertrand duopoly (no bundling): $\pi = 0$, no product differentiation
2. Bundling (partial differentiation): $\pi_i > 0$, service allocation distorted
3. First-best: Allocatively and productively efficient

Benefits:

- Bundling prevents Bertrand profit collapse
- Creates product variety (service-enhanced vs. basic)

Costs:

- Service provision misaligned with social cost
- Higher prices may reduce consumer surplus

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Summary and Policy Implications

Key takeaways (1)

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Bundling serves three distinct strategic purposes:

1. Price discrimination (monopoly)
   - Extracts surplus from heterogeneous consumers
   - Pure bundling often superior to separate selling
   - Mixed bundling maximizes profit
2. Entry deterrence (strategic)
   - Leverages existing market power into adjacent markets
   - Reduces entrant's residual profit below fixed costs
   - Exclusionary but may avoid cost duplication

Key takeaways (2)

3. Product differentiation (competition)
   - Softens Bertrand competition
   - Creates vertical differentiation via complementary services
   - Enables positive equilibrium profits

Antitrust implications

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Concerns:

- Bundling can facilitate monopolization (leverage problem)
- May reduce consumer choice and competition
- Entry deterrence is anticompetitive when exclusionary

Defenses:

- Efficiency gains: reduced advertising, integrated delivery, lower fixed costs
- Consumer benefits: discounted bundle prices, simplified purchasing
- Procompetitive: differentiation, overcomes Bertrand paradox

Principle: Analyze case-by-case. Effect depends on market power, cost structure, and intent.

Summary and next week

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Summary:

- Bundling exploits consumer heterogeneity: mixed bundling dominates, $\pi^{\text{mix}} > \pi^{\text{pure}} > \pi^{\text{sep}}$
- Pure bundling serves broader demand, yielding higher total welfare than separate selling
- Bundling deters entry by reducing the entrant's residual profit below fixed cost F
- Tying to a complementary service differentiates products and overcomes the Bertrand paradox
- Welfare effects are ambiguous: efficiency gains may offset anticompetitive exclusion

Next lecture: Lecture 8

Summary and next week

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Advertising

- When and why do firms advertise?
- Advertising as a signal of product quality
- Strategic use of advertising in oligopoly

References

Belleflamme, Paul, and Martin Peitz. 2015. Industrial Organization: Markets and Strategies. Cambridge University Press.

Shy, Oz. 1995. Industrial Organization: Theory and Applications. MIT Press.